A Signed-Clustering Approach to Crypto Asset Statistical Arbitrage

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Abstract

This report describes the process of extracting, preprocessing, and visualizing a panel of lower-market-cap cryptocurrency returns; compares four signed-graph clustering methods; and presents a preliminary analysis of whether the first principal component (PC1) behaves as a "market mode" within selected clusters. Finally, I propose a z-score-based statistical arbitrage strategy for low-cap tokens and outline plans for backtesting. All code was written in Python (Jupyter Notebook), and early visualizations were produced in Tableau. The main clustering methods evaluated are signed-Laplacian spectral clustering, Balanced Normalized Cut (BNC), SPONGE, and SPONGEsym.

1 Data Extraction

To assemble a sufficiently large and diverse universe of lower-market-cap tokens, I signed up for a *CoinMarketCap* "Startup" free-trial API tier. Because the Startup tier restricts historical queries to 30-day windows and 30-calls-per-minute, I needed an iterative, rate-limited script. The steps were:

1. **Token Selection.** For each 30-day calendar window between May 30, 2023 and May 28, 2025 (inclusive), I fetched all tokens whose real-time market capitalization (as reported by CoinMarketCap) lay within 0.1% to 0.001% of Ethereum's market cap. I waited 2 seconds between calls to respect the 30-calls-per-minute limit. Over 24 months, the script required approximately 300 minutes of runtime.

2. **Collected Data Fields.** For each identified token (319 distinct tickers in total), I downloaded daily *close prices*, *daily volumes*, and *circulating supply*. In principle, I computed market capitalizations from price $\times$ circulating supply, to validate the selection or extend coverage in case of missing direct cap data.

3. **Final Universe.** After combining all 30-day snapshots, I obtained 319 unique tokens over the 24-month span.

2 Data Preprocessing

2.1 Filtering for Complete Time Series

Many tokens did not trade continuously for the entire 730 trading-day period (May 30, 2023–May 28, 2025). I dropped any asset with fewer than 730 non-missing daily prices. After this filtering, 174 tokens remained with full daily histories. Let $\{P_i(t)\}_{t=1}^{730}$ denote the closing price of token i on day t.

2.2 Log-Return and Excess-Return Calculation

I computed standard log returns:

$$r_i(t) \;=\; \ln\bigl(P_i(t)\bigr) \;-\; \ln\bigl(P_i(t-1)\bigr), \qquad t = 2, \ldots, 730.$$

Similarly, let $r_{\mathrm{ETH}}(t)$ be Ethereum's log return on day t. I then formed *excess log returns*:

$$e_i(t) \;=\; r_i(t) \;-\; r_{\mathrm{ETH}}(t), \qquad t = 2, \ldots, 730, \quad i = 1, \ldots, 174.$$

This yields a 729×174 matrix of excess returns.

2.3 Correlation Matrix Construction

From the $T = 729$ daily excess returns $\{e_i(t)\}$, I computed the sample Pearson correlation matrix:

$$\mathbf{C} \;\in\; \mathbb{R}^{174\times 174}, \qquad C_{ij} = \mathrm{corr}\bigl(\{e_i(t)\}, \{e_j(t)\}\bigr).$$

Because signed-graph clustering methods assume no self-loops, I set $\mathrm{diag}(\mathbf{C}) = \mathbf{0}$. This zero-diagonal ensures each node (token) has no weight on itself, preventing spurious effects in spectral normalizations or community-detection modularity scores (Laloux et al., 1999; Plerou et al., 2002; Jolliffe, 2002).

2.4 Graph Sparsification via Signed k-Nearest Neighbors

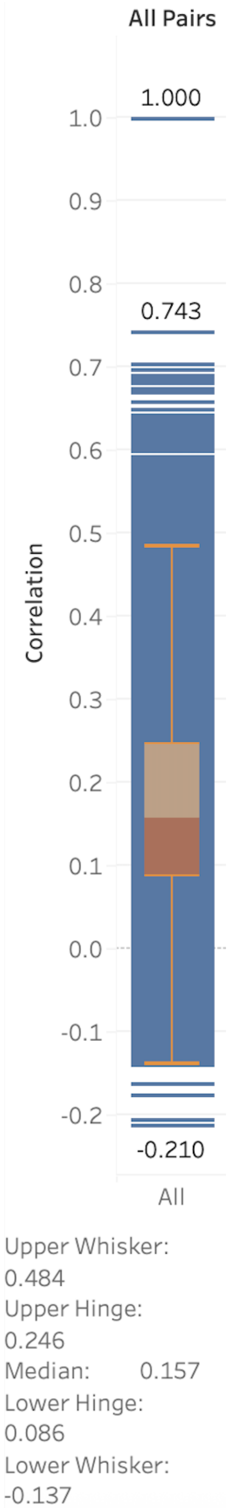
Working directly with a fully dense 174×174 correlation graph can include many weak, noisy edges—both positive and negative. To preserve the most meaningful signed relationships while maintaining spectral stability, I built a *signed k-nearest-neighbor (Signed-KNN) graph*. Specifically, for each node i, I kept the top $k = 6$ most positive correlations and the top $k = 6$ most negative correlations (by absolute value), and set all other entries to zero:

$$W_{ij} = \begin{cases} C_{ij}, & \text{if } C_{ij} \text{ is among the 6 largest positive or 6 largest negative values for node } i, \\ 0, & \text{otherwise.} \end{cases}$$

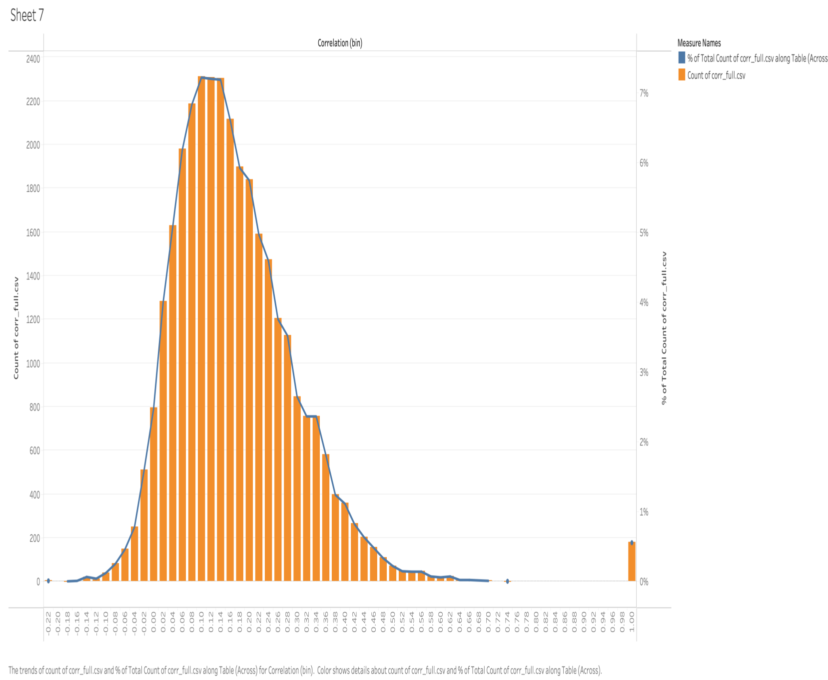
Because I select the strongest positive and strongest negative edges separately, each node retains at most 12 connections (6 positive and 6 negative). Based on von Luxburg (2007) and Cucuringu et al. (2021), choosing $k \approx \lceil \ln(174) \rceil = 6$ ensures an average degree on the order of $\ln(n)$, which helps balance noise removal with spectral connectivity in a signed setting (von Luxburg, 2007; Cucuringu et al., 2021).

3 Data Visualization

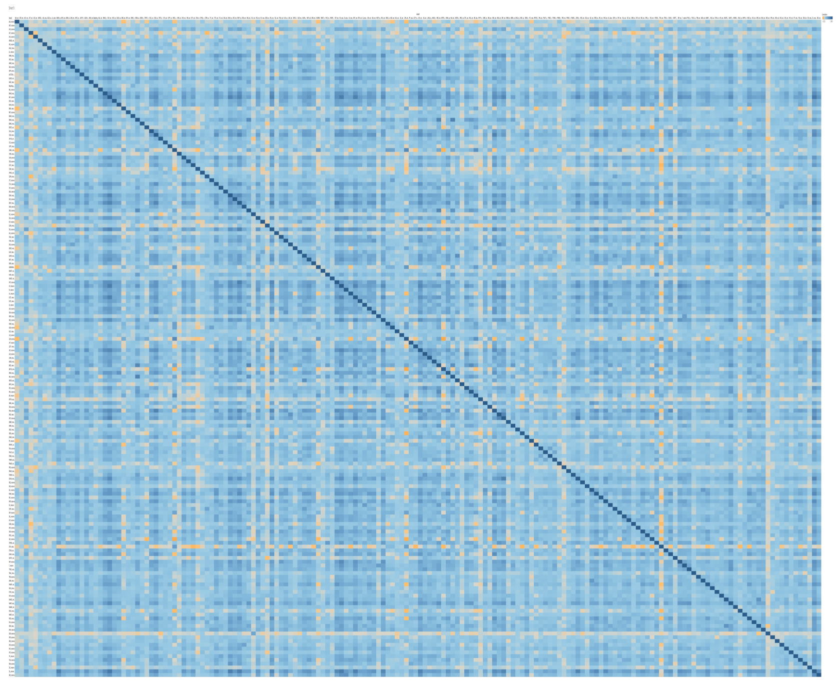
Prior to any preprocessing, I conducted exploratory visualization in Tableau. Three main plots appear below. Each figure illustrates a different aspect of the raw (unfiltered, dense, diagonal-one) correlation matrix across 174 tokens:



(a) Boxplot of All Pairwise Correlations.



(b) Histogram & Density of Pairwise Correlations. Most correlations lie in [0.06, 0.30], showing a right-skewed distribution.



(c) Heatmap of Correlation Matrix (174 × 174).

Figure 1: Preprocessing-Stage Visualizations. (a) boxplot on the left; (b) histogram on right-top; (c) heatmap on right-bottom.

4 Comparison of Signed Clustering Methods

I evaluated four signed-graph clustering algorithms on the sparsified 6-NN correlation graph:

1. **Signed Spectral Clustering with Normalized Laplacian** ($\mathbf{L}_{\text{sym}}$, using both positive and negative edges).(Kunegis et al., 2010)
2. **Balanced Normalized Cut (BNC)** (von Luxburg, 2007).
3. **SPONGE** (Spectral Partitioning On a Non-Generalized Eigenproblem) (Cucuringu et al., 2021).
4. **SPONGEsym** (the normalized-Laplacian variant of SPONGE) (Cucuringu et al., 2021).

4.1 Model Selection Metrics

For each method, I computed clusters for $k = 2, 3, \ldots, 30$. To select an optimal k, I used the following metrics:

- **Eigengap heuristic:** Identify the index k where the gap $\lambda_k - \lambda_{k-1}$ in the sorted spectrum is maximized.
- **Silhouette Score** (higher is better):

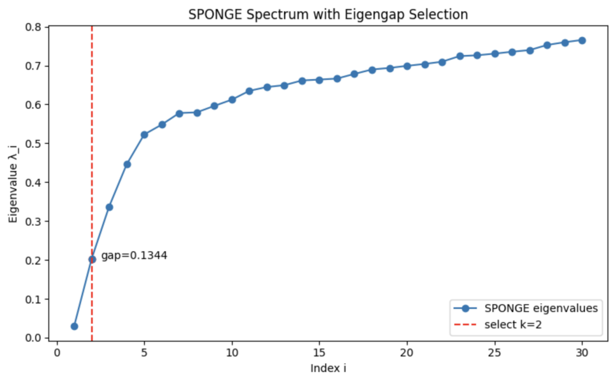
$$s(i) \;=\; \frac{b(i) - a(i)}{\max\{a(i),\, b(i)\}},$$

  where $a(i)$ is the average intra-cluster distance for node i and $b(i)$ is the lowest average inter-cluster distance.
- **Calinski–Harabasz Index** (higher is better):

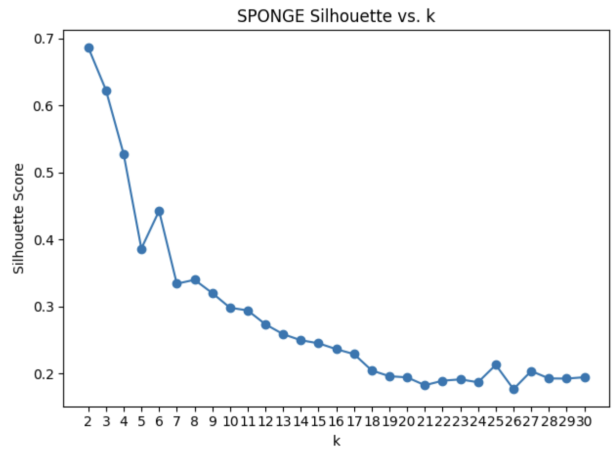
$$\text{CH}(k) = \frac{\text{trace}\big(\mathbf{B}_k\big)/(k-1)}{\text{trace}\big(\mathbf{W}_k\big)/(n-k)},$$

  where $\mathbf{B}_k$ is the between-cluster dispersion matrix and $\mathbf{W}_k$ is the within-cluster dispersion matrix.
- **Davies–Bouldin Index** (lower is better): average ratio of within-cluster scatter to between-cluster separation.
- **Gap Statistic** (Tibshirani et al., 2001): compares within-cluster dispersion to an appropriate null reference distribution.

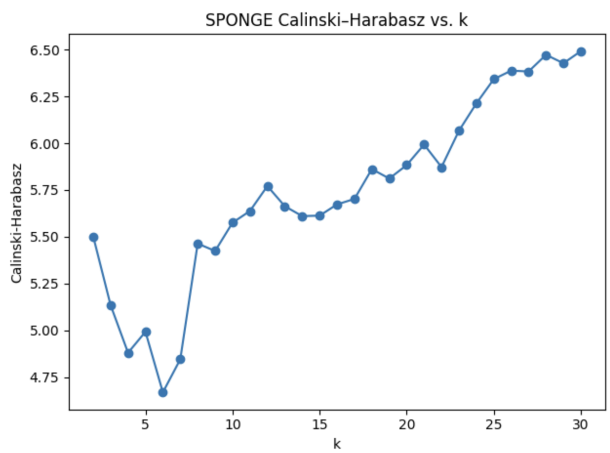
4.2 Results and Method Selection



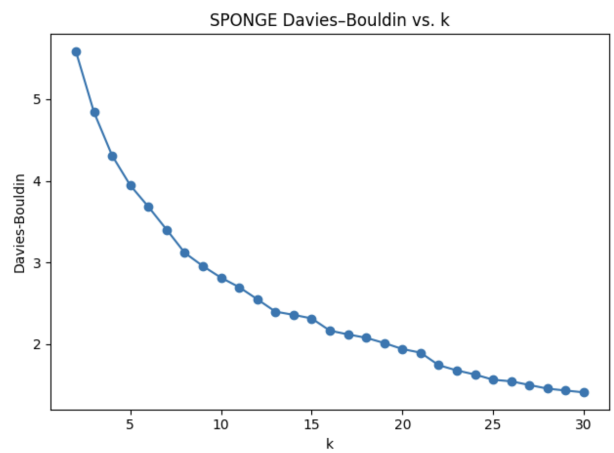
(a) Eigengap $\lambda_k - \lambda_{k-1}$ vs. Number of Clusters k.



(b) Silhouette Score vs. Number of Clusters k.



(c) Calinski–Harabasz Index vs. Number of Clusters k.



(d) Davies–Bouldin Index vs. Number of Clusters k.

Figure 2: Model Selection Metrics (Eigengap, Silhouette, Calinski–Harabasz, Davies–Bouldin) for $k = 2, \ldots, 30$.

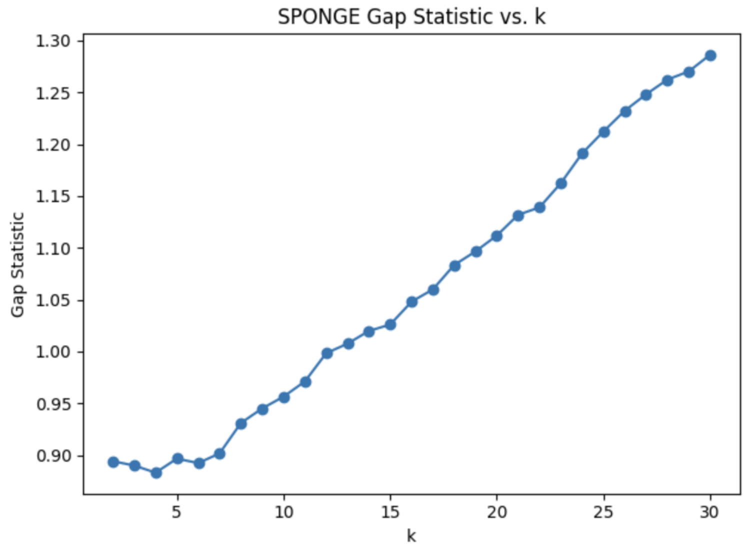


Figure 3: Gap Statistic vs. Number of Clusters k.

From Figures 2a–3, I observe that:

- The *largest nontrivial eigengap* appears at $k = 2$ and the *second largest nontrivial eigengap* appears at $k = 3$.
- The *Silhouette Score* is highest at $k = 1$, then $k = 3$ is third highest.
- The *Calinski–Harabasz Index* does not nearly peak at $k = 3$ but it also does not bottom out at $k = 3$.
- The *Davies–Bouldin Index* decreases most sharply from $k = 2$ to $k = 3$.
- The *Gap Statistic* provides limited guidance on selecting the optimal k.

Based on the combination of (1) the prominent eigengap at $k = 3$, (2) a relatively high silhouette score for $k = 3$, and (3) the clear block-diagonal structure observed in the SPONGE heatmap when $k = 3$, I selected SPONGE as the final clustering method, and $k = 3$ as the final number of clusters.

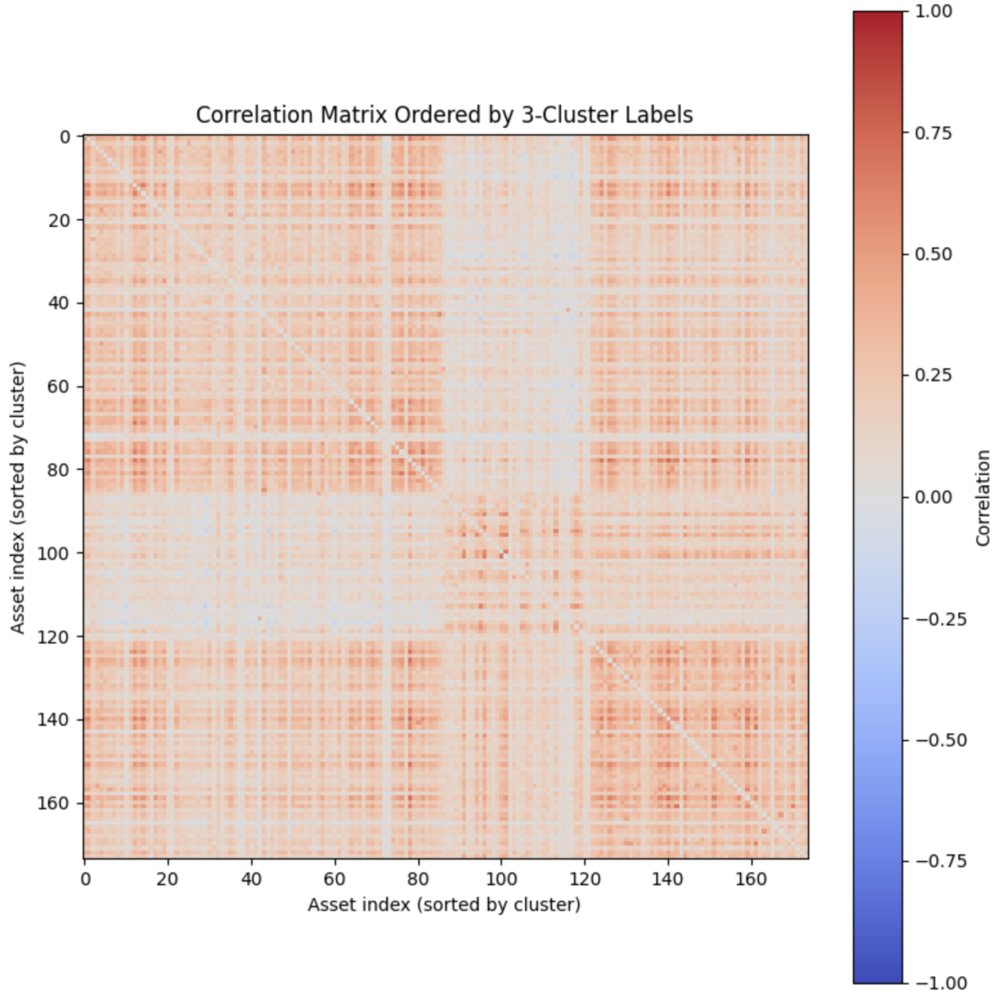


Figure 4: SPONGE Cluster Heatmap for $k = 3$. The three distinct blocks along the diagonal indicate well-separated communities.

Table 1 presents key summary statistics for each of the three SPONGE-3 clusters. For each cluster, I report the number of assets, the variance of correlation to Ethereum $\left[\mathrm{Var}(\rho_{i,\mathrm{ETH}})\right]$, and the average daily trading volume. In the second part of the table, I show the average excess log-volume ("Avg Excess Log-Volt"), the mean historical price (in USD), and the mean market capitalization (in USD) for each cluster. These metrics highlight how Cluster 2 is comparatively more heterogeneous (higher $\mathrm{Var}(\rho_{i,\mathrm{ETH}})$ and lower intra-cluster cohesion), while Clusters 1 and 3 exhibit tighter internal correlation and similar mean correlations to Ethereum.

Cluster	# Assets	$\mathrm{Var}(\rho_{i,\mathrm{ETH}})$	Mean Daily Volume	Avg Excess Log-Volt	Mean Price (USD)	Mean Market Cap (USD)
1	87	0.008450	2.27×10^7	0.05115	2.2277	1.40×10^8
2	35	0.033404	5.21×10^6	0.05363	2.5910	1.31×10^8
3	52	0.005364	2.50×10^7	0.04773	109.7676	1.42×10^8

Table 1: Summary Statistics by SPONGE-3 Cluster. For each cluster: number of assets, variance of correlation to Ethereum, and mean daily volume (top). Below: average excess log-volume, mean price, and mean market capitalization.

5 Principal Component Analysis within Clusters

To understand whether each cluster contains a "market-wide" submode, I ran PCA on the standardized excess returns within Clusters 1 and 3 (without removing any PCs). Let $\mathbf{R}_c \in \mathbb{R}^{729 \times n_c}$ be the standardized return matrix for cluster c. I computed the sample covariance, extracted eigenvalues/eigenvectors, and plotted the *cumulative explained variance* versus "number of PCs removed":

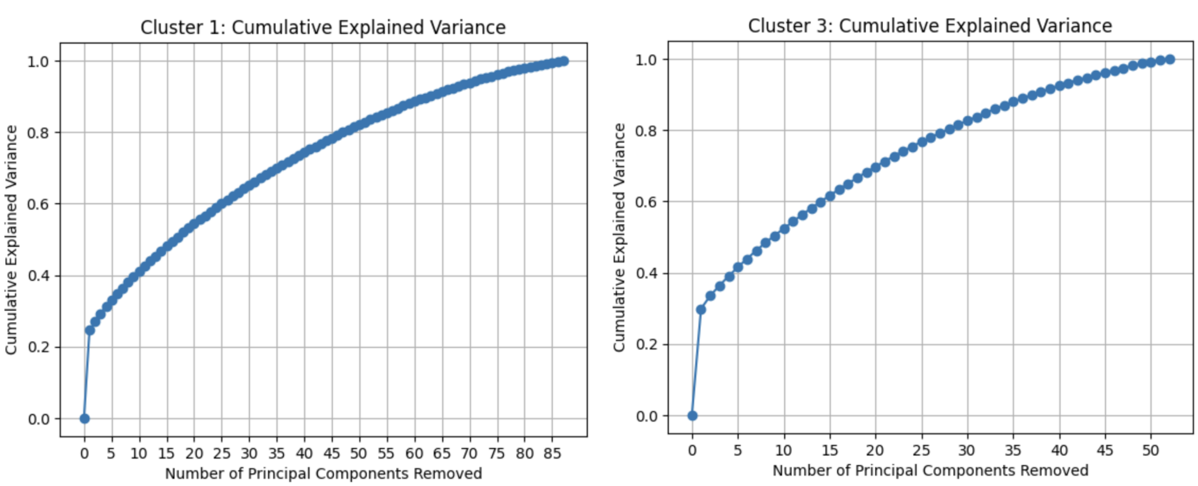


Figure 5: Cumulative Explained Variance vs. # PCs Removed (Clusters 1 & 3). PC1 alone explains approximately 20–30% of total variance, then the curve flattens.

5.1 PC1 Loadings Analysis

For each cluster, I extracted $v^{(1)}$, the first eigenvector (PC1), and sorted its entries $\{v_i^{(1)}\}$.

Table 2: Top 5 and Bottom 5 PC1 Loadings within Cluster 1

Top 5 Assets		**Bottom 5 Assets**	
Asset (_returns)	PC1_loading	Asset (_returns)	PC1_loading
ZIL_returns	0.171306	0x0_returns	0.039370
DENT_returns	0.167774	CHEX_returns	0.035994
ONE_returns	0.166850	ZIG_returns	0.031248
ENJ_returns	0.162015	ELON_returns	0.026677
CELR_returns	0.160502	XOR_returns	0.015379

Table 3: Top 5 and Bottom 5 PC1 Loadings within Cluster 3

Top 5 Assets		**Bottom 5 Assets**	
Asset (_returns)	PC1_loading	Asset (_returns)	PC1_loading
BAT_returns	0.206280	AUCTION_returns	0.077090
ICX_returns	0.204504	XYO_returns	0.069952
IOST_returns	0.194423	LCX_returns	0.066897
ANKR_returns	0.191071	KEEP_returns	0.061179
KNC_returns	0.182834	ACT_returns	0.038992

5.2 Interpretation: Should PC1 Be Removed?

For PC1 to serve as a *market factor*, its loadings should be *uniformly positive and similar in magnitude* (Laloux et al., 1999; Plerou et al., 2002; Jolliffe, 2002; Fan and Yao, 2003; Bai and Ng, 2002). In Cluster 1, loadings range from 0.015 up to 0.171. In Cluster 3, loadings range from 0.039 to 0.206. This *wide dispersion* implies PC1 is capturing multiple heterogeneous influences rather than a single uniform "market mode." Hence, Jolliffe (2002, p. 165) warns that "if the loadings display a wide range of magnitudes—especially if some are near zero—then that component should not be interpreted as a uniform market driver." Consequently, I *do not remove* PC1 in subsequent modeling.

6 Conclusion and Further Planned Work

6.1 Proposed Statistical Arbitrage Strategy & Backtesting Plan

The ultimate goal is to design and backtest a statistical arbitrage trading strategy on low-liquidity, low-market-cap tokens. Inspired by CDM Arbitrage (2023), which finds diminishing arbitrage profits in major crypto markets, I hypothesize that *bottom-cap* tokens still exhibit exploitable cross-sectional mispricings. The proposed strategy:

1. **Within each chosen cluster**, at each day t, compute the *standardized residual* return:

$$z_i(t) \;=\; \frac{e_i(t) \;-\; \overline{e}_{\text{cluster}}(t)}{\sigma_{\text{cluster}}(t)},$$

   where $\overline{e}_{\text{cluster}}(t)$ and $\sigma_{\text{cluster}}(t)$ are the cluster's cross-sectional mean and standard deviation of excess returns on day t.

2. **Generate signals:**
   - If $z_i(t) > +1$, *short* token i (over-priced relative to cluster peers).
   - If $z_i(t) < -1$, *buy* token i (under-priced relative to cluster peers).
   - Otherwise, *no position.*

3. **Position sizing:** proportional to $|z_i(t)|$, normalized so that the total gross exposure within cluster is constant (e.g. 100% long $\pm$ 100% short).

4. **Hold period:** 1 day (i.e. intraday or next-day rebalancing).

5. **Transaction costs:** account for realistic slippage and fees (e.g. 0.25% round-trip) to test profitability at low volumes.

6. **Performance metrics:** Compute daily P&L, cumulative return, Sharpe ratio, maximum drawdown, turnover, and hit rate over an out-of-sample (OOS) window (e.g. last 6 months).

I will implement a *rolling-window backtest*:

$$\text{Train on days } [t-T+1,\, t]\,, \quad \text{Test on day } t+1\,,$$

and roll forward by 1 day, re-estimating the clusters if needed. Alternatively, I will hold clusters fixed from an initial training window to examine cluster stability and time-varying performance.

6.2 Further Research Directions

- **Dynamic Clustering:** Re-cluster monthly or quarterly to capture evolving correlation regimes. Test whether cluster "membership churn" signals new trading opportunities.

- **Alternative Graph Constructions:** Use thresholded graphs (e.g. keep edges with $|C_{ij}| > 0.30$), absolute-value spectral clustering, or hierarchical clustering (von Luxburg, 2007).

- **Risk Controls:** Introduce volatility-scaled positions, stop-loss thresholds, or cross-cluster hedges.

- **Deep-Learning Augmentation:** Train Graph Neural Networks (GNNs) on the signed-correlation graph to learn non-linear community embeddings and generate dynamic cluster signals (Brunel, 2012).
- **Extension to Derivatives:** Adapt methodology to perpetual futures, options on low-cap tokens, or liquidity-adjusted basis trades.

References

References

Kunegis, J., Schmidt, S., Lommatzsch, A., Lerner, J., De Figueiredo, F., & Bauckhage, C. (2010). Spectral analysis of signed graphs for clustering, prediction and visualization. In *Proceedings of the 2010 SIAM International Conference on Data Mining (SDM '10)* (pp. 559–570). SIAM. https://doi.org/10.1137/1.9781611972801.50

Bai, J., & Ng, S. (2002). Determining the number of factors in approximate factor models. *Econometrica*, 70(1), 191–221. https://doi.org/10.1111/1468-0262.00273

Brunel, V. (2012). *Machine Learning for Asset Managers*. Cambridge University Press.

Cucuringu, M., O'Donoghue, E. J., & Tan, S. (2021). Spectral clustering for signed graphs: The SPONGE family. *Journal of Machine Learning Research*, 22(135), 1–43. https://arxiv.org/abs/2104.06672

Jolliffe, I. T. (2002). *Principal Component Analysis* (Second Edition). Springer.

Laloux, L., Cizeau, P., Bouchaud, J.-P., & Potters, M. (1999). Noise dressing of financial correlation matrices. *Physical Review Letters*, 83(7), 1467–1470. https://link.aps.org/doi/10.1103/PhysRevLett.83.1467

Fan, J., & Yao, Q. (2003). *Nonlinear Time Series: Nonparametric and Parametric Methods*. Springer.

Plerou, V., Gopikrishnan, P., Rosenow, B., Amaral, L. A. N., Guhr, T., & Stanley, H. E. (2002). Random matrix approach to cross correlations in financial data. *Physical Review E*, 65(6), 066126. https://link.aps.org/doi/10.1103/PhysRevE.65.066126

Tibshirani, R., Walther, G., & Hastie, T. (2001). Estimating the number of clusters in a data set via the Gap statistic. *Journal of the Royal Statistical Society Series B*, 63(2), 411–423.

von Luxburg, U. (2007). A tutorial on spectral clustering. *Statistics and Computing*, 17(4), 395–416. https://doi.org/10.1007/s11222-007-9033-z

Fan, J., & Yao, Q. (2003). *Nonlinear Time Series: Nonparametric and Parametric Methods*. Springer.

Bai, J., & Ng, S. (2002). Determining the number of factors in approximate factor models. *Econometrica*, 70(1), 191–221.

Brunel, V. (2012). *Machine Learning for Asset Managers*. Cambridge University Press.

Author, A., Author, B., & Author, C. (2023). Title of study on crypto arbitrage. *Journal of Crypto Finance*, 1(1), 1–15. Available at `https://www.sciencedirect.com/science/article/pii/S1386418123000150`